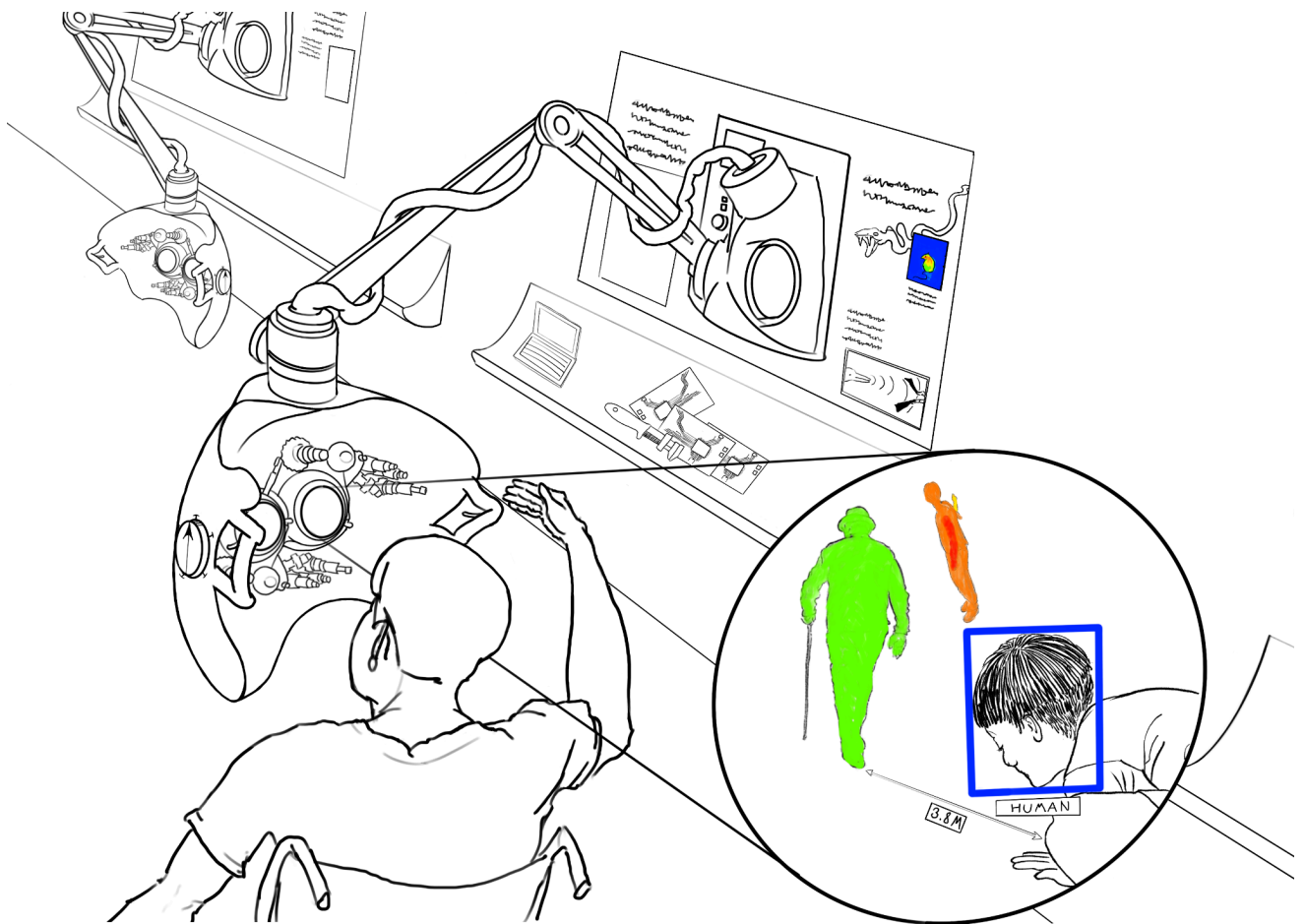


Exhibit Code and Name:	3.2.4 Eye Am Robot
Zone:	03 Cyborgs
Subzone:	3.2 Swiss Army Appendage
Version: 1	Revision Number / Date: Rev __, YY-MM-DD

Robot vision:



1. Exhibit Type	A
2. Exhibit Description	A series of fixed, wall mounted AR viewing stations enable visitors to see the world the way a robot does - in first person view.

	A physical dial on the side of each viewing station apparatus shifts the visitor between different vision modes. Each mode experiences alternative ways in which machines see the world, such as infrared sensors, X-rays, LiDAR, Radar and neural network based object detection. Corresponding contextual information explains how robots convert data into vision through each viewing station mode. Text and graphical elements explore the diverse systems of vision present across Earth's animal kingdom, and explain how machine vision has developed in response. This exhibit examines how organic and robotic beings use data in the creation of 'vision', emphasising the complexity and variety of what we understand as sight.
3. Interactive Description	Eye Am Robot • Visitors are invited to interact with a set of AR headsets that are fixed in a line. • Each headset has the ability to swivel (up and down, back and forth) to view different angles of a scene within the Zone, and can be height adjusted, but otherwise cannot be moved from their fixed area. • Each headset provides the opportunity for visitors to 'see like a robot'. Handles are shaped onto both sides of each AR mask for ease of motion, and sanitary wipes are nearby as well for ease of cleaning before/after use. • When a visitor first experiences robot vision, internal sensors will reset the experience from the last visitor, ensuring it will begin with the aesthetically foreign image segmentation and object detection vision. ◦ Images will not appear very focussed to begin with, blunt silhouettes and plain colour differentiation override objects in the viewport ◦ As they focus in on the scene, visitors will see calculations begin in real-time, labelling and identifying objects and evaluating the distance between them.. ◦ As the visitor turns an overstated dial on the side of the viewer the form of vision switches, allowing the visitor to move from the initial object detection vision through to X-ray, infrared, LiDAR and Radar visions respectively. • There will be additional contextual information in the form of AR text and auditory narration which will explain what the visitor is experiencing with each vision type.

- Neural network object detection, image segmentation: the first vision type that the AR viewers explore utilises machine learning algorithms to generate real time blob tracking and object detection on the surrounding people and environment. Image segmentation processes first analyse the pixels in an image to find distinct regions. Blob tracking then finds these connected regions and tracks or maintains their shape, while object detection uses machine learning to identify these 'blobs', assessing what the shapes are most likely to be. Once identified, objects can be assigned a size, and then distance between objects can be calculated. While these processes happen quietly and quickly in machine vision applications, we can slow down and visualise space at each stage for visitors to better understand or get a feel for these processes. This vision type is the introduction to machine vision because it is the closest to human vision, with camera lenses instead of human lenses, and silicon software instead of grey-matter wetware all to identify, label, measure, sort and remember what we see.
 - "Pixel by pixel, algorithmic processes allow robots to make sense of images. First assessing probabilities to make groups and colour values to hold their shapes, then comparing these across hundreds of thousands of images in a 'neural network', machines can 'see' what is around them, at what size, and at what distance."
- LiDAR: through emitting light waves and measuring the distances they travel the LiDAR system creates a high detail 'point cloud' render of the space. The short length of these light waves mean that they reflect off of smaller particles, providing much more feedback to create more detailed and complex 3D visualisations with. LiDAR scans can be pulsed at a regular rhythm to continually create point clouds of the visitors surroundings, creating an intriguing impression of moving objects and people.
 - "LiDAR uses lasers to visualise three dimensional space through measuring how long it takes for the emitted light to reflect off another particle back to its source, placing a digital point at each reflected location. Light waves' short length means that they reflect back from contact with

almost any particle, creating a dense 'cloud' of points detailing space in LiDAR visualisations."

- Radar: radio waves are emitted to create simple point clouds detailing the visitors surroundings in real time. Through emitting radio waves and measuring the time that they take to reflect back to their source, as well as their strength and direction, an image of space is formed. Radio waves are unique in that their long wave-length can pass through non-conductive materials such as plastic, glass or wood, and conditions such as fog or clouds. Radar is great at detecting motion and speed (doppler effect). In this instance, the radio waves can also be used to measure the heartbeat of the visitor, which will be displayed back to them visually in real time.
 - *"The long length of radio waves used in radar vision means that less information is reflected and measured. Although similar methods to LiDAR are used to 'see', a robot has a lot less thinking to do with each Radar scan! This allows radar to be effective at measuring movement, as less information to process means pulses of radio waves can be emitted at a very high rate. These pulses create a map of even very fine movements, such as your heartbeat (heartrate popup appears)."*
- Infrared: infrared can be directly experienced through the use of a thermal camera in the AR headset. Infrared vision works through measuring heat and transforming the measurements into digital data, accurately mapping objects and beings through their heat levels. Perfect for examining the difference between warmer humans and colder robots in the zone!
 - *"Very hot objects like the sun or the stars emit light visible to humans, but humans, being much less hot, emit a different kind of light. This is called Infrared light, or Infrared radiation, and cannot be seen by your human eye, instead only being felt as heat. Robots however can detect and measure Infrared, allowing them to see in complete darkness!"*

- X-ray: x-ray vision cannot be used in real time in the space, but it can be spoofed with certain objects in the space having an X-ray like overlay applied to them, with visual cues to steer the visitors attention toward these particular objects. These would include seeing through aluminium workshop type electronics to see their motherboards. This action can capture x-ray visuals mapped on a series of pre-determined, pre-mapped objects as well as on the visitors hands, showing a variety of common use cases for this form of vision.
 - “Traditional medical X-ray imaging requires the X-rays to pass through an object to a detector on the other side to generate accurate images, why don’t you try taking something from the shelf and holding it out between you and the detector on the wall above it?”

Contextual information surrounding interactive:

- Unpack the sensors that develop vision beyond camera view, such as Infrared. What other inputs are fed into computers? And how is this data converted into ‘vision’.
- What is the complex data coming in and how is it converted into information that the robots can use?
- Unpack the sensors that help the robot to visualise space and distance in vision systems like LiDAR, radar and sonar.
 - How is the data on space/distance converted into information that the robots can use to be aware of their surroundings?
- Videos and diagrams of animals with different systems of vision which have inspired the development of robot vision.
 - This could show the animals within their natural environment to showcase the adaptive qualities of their particular vision type. E.g:
 - Snakes hunting for prey with thermal vision
 - dolphins using echolocation to ‘see’ through murky or darker waters
 - Bees seeing ultraviolet light to find UV light emitting flowers

	Diagrams or animations of how humans see/how data is processed by the human brain - and how this may differ from other animals (e.g. with more cone types, or colour receptors) and robots.
4. Learning and Emotional Objectives	Learning Objectives - Encourage visitors to reconceptualise their own human brains as immense visual data processing centres. - Understand that human vision is only one of many possible ways of perceiving the world. - Appreciate the diverse ways animals see their environments and how these adaptations support survival and evolution. - Understand how data is converted into information that is utilised by robots in order to engage with their environments. Sci-Fi Objectives - Explore the science-fiction trope of enhanced and non-human perception through robotic and alien viewpoints. - Consider how future robots and artificial intelligences might perceive the world differently from humans. Emotional Objectives - Inspire awe at the huge number of different ways to see the world, and the advantages and disadvantages of human sight. - Have fun experiencing the different ways visual data can be processed. - Understanding and empathising with the robot experience.
5. Theming Notes	This exhibit will be placed within a section of the cyborg workshop, seemingly dedicated to the development of robotic heads, or masks. Tools and spare parts are abundant, as well as more industrial - workshop type embellishments.
6. Accessibility Notes:	Integrated Design Considerations for; Physically impaired: Hearing impaired: Vision impaired:
7. Reference links and	Interactivity and Exhibit References:

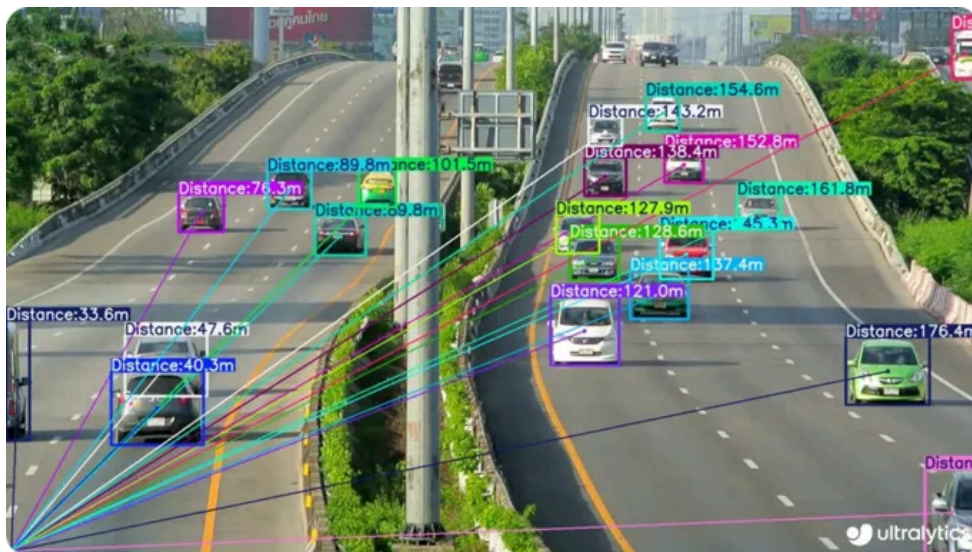
images	• Text: Content References: • Text:
8. Source	Custom Fabricated
9. Key Components	Concept Phase • AV Hardware ◦ Text • Physical Build Elements Schematic Phase • See Schematic Drawings for Build Elements • See AV Hardware Schedule
10. Size	Concept Phase • Estimated Size Schematic Phase • See Schematic Drawings
11. Estimated Cost	See Price Schedule at Schematic Phase
12. Operational Requirements	To be developed at Schematic stage of the project Safety: Text Modularity / Removability: Text
13. Maintenance Notes	To be developed at Schematic stage of the project • Text

3.2.4 Eye Am Robot (Previously 4.15.1)

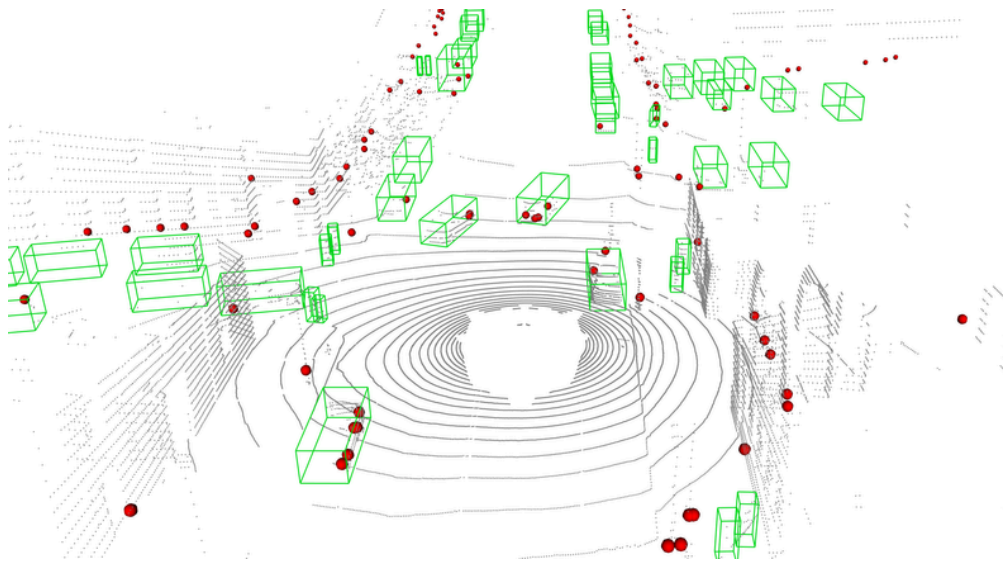
3.2.3 Eye Am Robot scratchpad

A	Consider creative ways to prompt the visitor to re-conceptualize their own human brains as immense visual data processing centres. Demonstrate how aspects of human vision, insect vision, and other organism's ways of sensing the world have influenced computer vision. Consider VR/AR experiences where visitors can be put in the (metaphorical) shoes of an android and experience the local environment through the eyes of a robot.
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Distance measure with object detection:



Radar point cloud:



LiDAR point clouds:

